

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
2 HT	(a)	(i)		$\frac{3.2}{2}$ or $\left(\frac{3.2}{8}\right) \times 4$ or $2.8 - 1.2$ (1) $X = 1.6$ [cm] (1)		2		2	2	
		(ii)		Substitution into: $\text{speed} = \frac{\text{distance}}{\text{time}} = \frac{40}{5}$ (1) $= 8$ [cm/s] (1) N.B. Substitution into wave equation i.e. $f = \frac{v}{\lambda} = \frac{40}{5} = 8$ don't award the first 2 marks Manipulation and substitution of $v = f\lambda$ i.e. $f = \frac{8(\text{ecf})}{1.6(\text{ecf from part(i)})}$ (1) $= 5$ [Hz] (1) If no workings shown answer of 8 on answer line award no marks. If intermediate calculation shown with 8 on the answer line award the first 2 marks. Alternative: $= \frac{40}{1.6(\text{ecf})}$ (1) = 25 waves (1) [in 5 seconds] Frequency = $\frac{25}{5(\text{ecf})}$ (1) = 5 [Hz] (1)	1	1		4	4	